

E28F002BXT@TSOP40(1).BIN analysis report

Executive identification

- Chip/dump size: Intel E28F002BXT, 2-Mbit / 256 KiB boot-block flash; file size 262,144 bytes / 0x40000.
- Likely role: IBM Palm Top PC110 main/system BIOS flash image.
- Confidence: Very high.
- SHA-256: 232101c88466f311bcc32fbc215a4d7569f695ce19f9c07ca67ce2aee5232312
- MD5: 6de4281a58509438a2773365ba6b1371
- Byte entropy: 6.8383 bits/byte.

Evidence from this dump

- Starts with a PC-style 0x55 0xAA ROM signature and an IBM copyright/date string.
- Contains embedded strings: "IBM VGA Compatible BIOS", "Chips 65535 VGA 32KB BIOS", "Version 2.0.2", "APM BIOS 1.00.27", "RIOS", "39H4551", and IBM copyright/date strings.
- Offset 0x3FFF0 contains the real-mode reset tail EA 5B E0 00 F0, a far jump to F000:E05B when mapped at the top of the PC BIOS address space.
- Hashes match a public PC110 BIOS dump inventory, which makes the identification very strong.
- Structure is mixed: code, headers, strings, blank fill, and probably compressed/packed or table-heavy blocks. It should not be treated as one linear x86 program.

Structural notes

- Active 4 KiB ranges (heuristic, excludes mostly 0x00/0xFF blocks): 0x00000-0x17FFF, 0x20000-0x3FFFF
- Longest 0xFF runs: 0x17721 len 35038, 0x2FB0C len 20, 0x2FB2C len 20, 0x325AC len 20, 0x27224 len 16
- Longest 0x00 runs: 0x3F0FC len 1861, 0x3EC63 len 756, 0x3E067 len 604, 0x3F85C len 538, 0x3E838 len 335
- PC option-ROM signatures found:
 - 0x000000: size byte 0x00 (0 bytes), checksum mod 256 None, valid=None, header 55 aa 00 eb 45 90 30 39
 - 0x020000: size byte 0x47 (36352 bytes), checksum mod 256 117, valid=False, header 55 aa 47 eb 3f 90 00 00
 - 0x026F84: size byte 0x55 (43520 bytes), checksum mod 256 217, valid=False, header 55 aa 55 aa 55 aa 55 aa
 - 0x026F86: size byte 0x55 (43520 bytes), checksum mod 256 114, valid=False, header 55 aa 55 aa 55 aa 55 aa
 - 0x026F88: size byte 0x55 (43520 bytes), checksum mod 256 11, valid=False, header 55 aa 55 aa 55 aa 55 aa
 - 0x026F8A: size byte 0x55 (43520 bytes), checksum mod 256 164, valid=False, header 55 aa 55 aa 55 aa 55 aa
 - 0x026F8C: size byte 0x55 (43520 bytes), checksum mod 256 61, valid=False, header 55 aa 55 aa 55 aa 55 aa
 - 0x026F8E: size byte 0x55 (43520 bytes), checksum mod 256 214, valid=False, header 55 aa 55 aa 55 aa dd 77

Interesting strings

- 0x000006: 09/19/95'(C) Copyright IBM Corp. 1994, All Rights Reserved.
- 0x02001E: IBM VGA Compatible BIOS. 4
- 0x02008B: a8Chips 65535 VGA 32KB BIOS
- 0x0200A8: Version 2.0.2
- 0x02012B: Copyright (C) 1994 Chips and Technologies, Inc. All Rights Reserved.
- 0x021FDF: %\$&U&CHIPS 65535 Flat Panel VGA

- 0x02901E: APM BIOS 1.00.27
- 0x029030: (C) COPYRIGHT RIOS SYSTEMS, 1993, 1994. ALL RIGHTS RESERVED.
- 0x030000: 39H4551 (C) COPYRIGHT IBM CORPORATION 1981, 1995 ALL RIGHTS RESERVED 11/08/953
- 0x03230A: Overflow
- 0x032438: Line buffer overflow
- 0x0324AF: FIELD overflow
- 0x03712C: TOKEN RING
- 0x03713C: IBM Corp.
- 0x037146: EthernetPSR
- 0x03E000: C0PR. IBM 1981, 1995

Included supplemental files

- strings/E28F002BXT_PC110_BIOS.strings.txt - all printable strings found by a simple ASCII scanner.
- hexdumps/E28F002BXT_PC110_BIOS_selected_hexdumps.txt - selected annotated hexdumps.
- disassembly/E28F002BXT_PC110_BIOS_key_regions_i8086_disasm.txt

Caveats

- This is static triage, not a full reverse-engineering project. Hardware register names, MCU peripheral vector names, and exact memory maps require board schematics, data books, or emulator instrumentation.